



Ayni Mesh

Frontier open models on hardware you already own.

The judgment stage of the Ayni marketplace: a swarm of community desktops serving 7B to 700B open models as one model, metered, verified and paid.

Built on Colibri and Lumabri (Apache 2.0) · Ayni 1.0 keeps running unchanged

1.0 taught us where the value stops.

Ayni 1.0 runs a 0.5B model on phones and laptops. It is cheap, sealed per device, and the supply is enormous. It does mechanical work well and judgment work not at all.

\$0.027

per 1M tokens for a 226-token log line on the 0.5B class; 3 to 5x under the nano tier

2 of 5

distinct answers the 0.5B model gave across 12 labelled security lines: not a classifier

0

devices in the 7B to 700B classes online today; the rate card prices them, nobody serves them

2 stages

phones normalize and label millions of items; something else must judge the flagged subset



What phones sell

Extraction, closed-set labelling, embeddings, redundancy votes. Mechanical, short, constrained, enormous. Keep it as it is.



What nobody sells yet

"Is this an incident?" needs a model that reasons. Hosted APIs sell it with their retention policy attached. Ayni has no supply for it.

Two open projects moved the wall.

Colibri · the engine

- Pure C, no dependencies. Nine frontier MoE families, one command line: GLM-5.3 (744B), DeepSeek V4 Flash (284B), Kimi K3 (2.8T), Qwen3.6 (35B), OLMoE (7B).
- Treats VRAM, RAM and SSD as one memory hierarchy; streams only the experts a token routes to.
- GLM-5.3-Flash (321B) on a 25 GB desktop. 1.8 tok/s on a 128 GB CPU box; 6.8 tok/s on six RTX 5090s.
- Placement may change speed, never semantics: outputs validated token-exact.
- 30,000 stars, daily commits, Apache 2.0.

Lumabri · the swarm

- Same author. Runs one model as a chain of layer segments or expert peers over a tracker.
- Signed weights, Ed25519 identities, ChaCha20-Poly1305 transport, NAT relay through the donor's own outbound tunnel: nothing to configure at home.
- Leases, fencing, snapshot and restore: a chat survives losing a peer. N% of calls re-executed on another peer and compared byte for byte.
- Measured: the network changed 0 of 24 tokens; 6 tok/s on a LAN, 1.1 tok/s at 30 ms internet RTT.
- No economics, no attestation, no phones, no Windows yet.

Mesh adds the layer the swarm is missing.



Engine (Colibri)

- Model math, formats, kernels
- CPU, Metal, CUDA, Vulkan tiers
- Pinned upstream, not forked



Swarm (Lumabri)

- Discovery, NAT relay, leases
- Segments and expert peers
- Verification, failover, snapshots

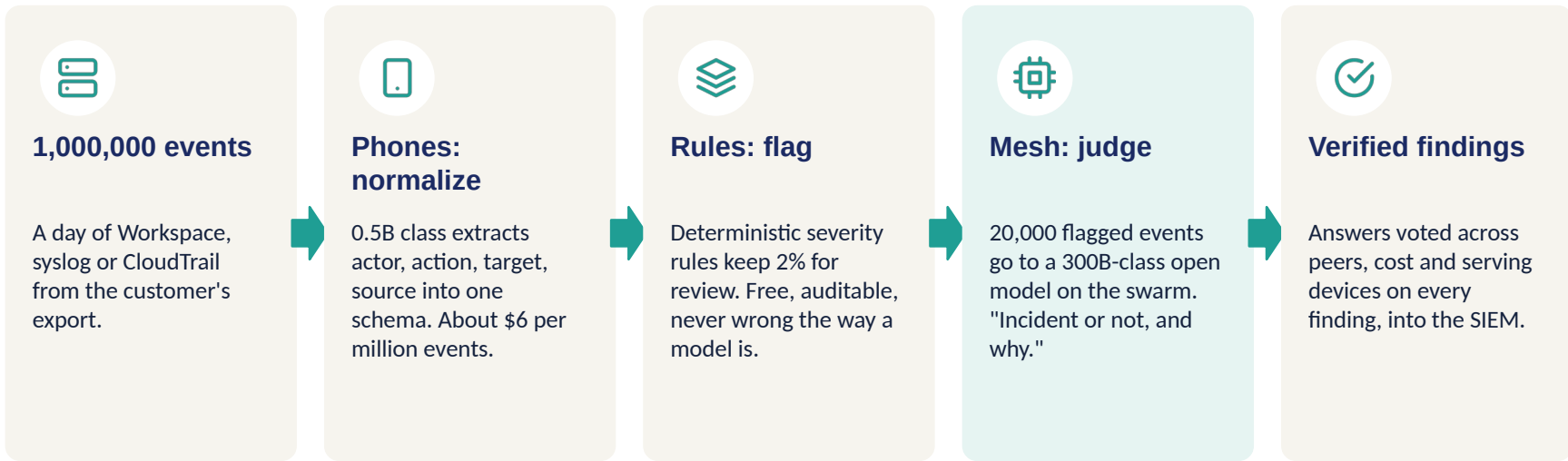


Marketplace (Ayni)

- Who may be a peer: attestation tiers
- What it costs: quotes, metered earnings per peer
- What runs where: price, RTT, benchmark
- Council on shape, never content

Same buyer surface as 1.0. Quote, accept, run, collect: the API, the Python client and the MCP server do not change. A Mesh route is one more provider class in the scheduler, priced per million tokens like everything else.

Phones do the million, Mesh does the thousand.



98%

of the volume never touches a large model; it is priced at the phone rate

2%

reaches the judgment class, where price per correct answer is the metric

1 API

one quote for both stages; MCP tools unchanged; per million tokens

Measured this week, not promised.

Test	Where	Result
Expert-peer swarm, tiny fixture, 4 peers	GCP e2-standard-8, loopback	1,984 remote expert calls, 8.4 ms per layer round, tokens identical to the local engine, incompatible build refused, spot-check verification agreed
Two-peer layer chain, real OLMoE-1B-7B	GCP, 8 CPU cores, no GPU	decode 12 to 13 tok/s, prefill 4 to 5.5 s, 3.8 MB on the wire for 48 tokens, generated ids identical to local Colibri
WAN chain, real OLMoE, 87 ms round trip	peers on GCP, chatter on a 16 GB Mac; then Mac as relay-only peer behind home NAT	3.5 to 4.6 tok/s, about one round trip per hop per token; faster than the Mac running the model alone (1.6 to 2.6 tok/s)
Model conversion	GCP	OLMoE 13.8 GB checkpoint to a 7 GB int8 container in 3.5 minutes, shard by shard
Two-region chain, real OLMoE, us-east1 to europe-west1, 107 ms RTT	GCP, two VMs	decode 4.1 to 4.7 tok/s (one WAN hop per token), prefill 9 s; symmetric from either side; tracker refuses unreachable peers
Lumabri's own release gate	author's runs	network changed 0 of 24 tokens; LAN 6.0 tok/s vs internet 1.1 tok/s vs slow-disk local 0.04 tok/s

The engine and swarm are correct, and a laptop renting two peers across the internet beat itself. The remaining gap to same-host speed (12 tok/s) is the price of the WAN; batching amortises it, and that is the next measurement.

What Mesh sells, and what it does not.



Sovereignty

Any open model on hardware you own or a community you choose. Pick peers by country. Nothing leaves the machines you selected.



Model choice

Every open checkpoint Colibri runs, the day it is released, unfiltered and unchanged. No vendor deprecates your model.



Verified answers

Replica re-execution and majority votes across peers, attested tiers, Council on shape. A finding carries who served it and what it cost.



Batch price on idle hardware

Marginal cost of an idle desktop is near zero. For overnight judgment work, price per correct answer competes with hosted 70B, once batching is proven. We publish the measurement, not the hope.

Not for sale

- Sub-second interactive chat across the internet. Sequential layers pay a round trip each.
- Cheaper tokens than an H100 for models that fit an H100. Small models belong on phones and GPUs.
- Prompt secrecy at the community tier. Peers see activations; use attested or private pools.

Three wedges, in this order.

1 · Private Mesh

Bring your own desktops

- A team's 5 to 50 machines become one 300B-class model overnight, with Ayni's scheduler, metering, attestation and audit.
- Sold as software plus coordination: per seat or per machine per month, plus metered tokens.
- No supply to bootstrap; the customer owns the hardware. The LAN-router crowd (NVIDIA PAIR users) is the first list.

2 · Community Mesh

The public swarm

- Desktop and gaming-PC owners with NVMe join as peers and earn per segment or expert call.
- Buyers quote per million tokens by class, choose tier and country, get verified answers.
- Opens when three-donor swarms beat one machine by 2x on the release gate.

3 · Overflow

Private first, community when full

- A private mesh spills its queue to the community swarm at a quoted price when its own machines are saturated.
- The same route on both sides; the buyer sets the boundary with a trust tier.
- This is the 1.0 hybrid marketplace story, now with a judgment class to sell.

Who buys first, and who lends the machines.

Buyer	Workload	Why Mesh
Security teams and MSSPs	Nightly judgment on flagged events; incident write-ups	Data stays on chosen machines; verified findings; open models they can audit
Regulated SMBs, EU and LatAm	Document review, contract triage, compliance checks	Residency by peer selection; no US-hosted API in the loop
Data-ops and labelling shops	Silver labels, quality checks on the phone stage's output	Batch price on idle hardware; redundancy votes as QA
Research groups	Evaluation sweeps on open frontier models	Any checkpoint, no rate limits, pay per token to a commons



Supply: desktops, not phones

Gaming PCs and workstations with 32 GB+ RAM and an NVMe drive; small teams' LAN clusters as one peer; laptops overnight. Linux and macOS now; Windows is the gap to close.

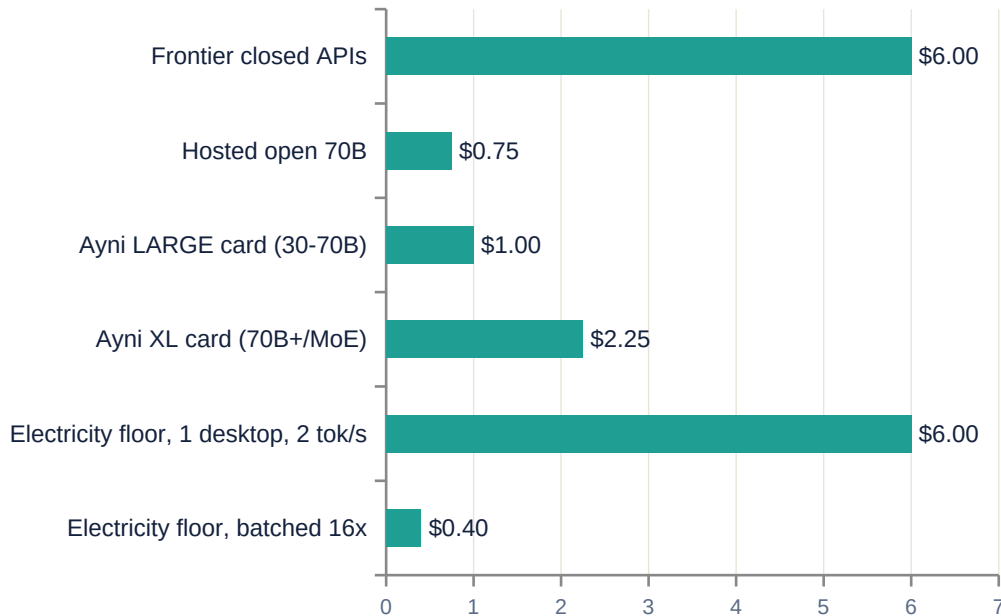


Earnings that mean something

A desktop serving a 300B-class segment earns at the LARGE rate, not the phone rate. Overnight donation pays for the electricity and then some; we publish the measured number.

Per million tokens, with the cost floor shown.

Approximate list prices and cost floors, USD per 1M tokens



- Every Mesh price is per million tokens by class, like 1.0 and like every vendor. The LARGE and XL cards already exist.
- The floor is electricity, not hardware: a 200 W desktop at 2 tok/s burns about \$6 of power per million tokens. Batching sixteen sequences through the same expert loads is what brings it under \$0.50. Colibri's batched expert unions and Lumabri's 8-client gate measure exactly this.
- So the honest position: for single-stream frontier tokens Mesh is not cheaper than a data center. For batched, overnight judgment on open models it can be, and we will publish the run before we sell it.
- Provider share 70%, attested 1.4x, confidential 3x, spot 60%: unchanged from 1.0.

What can go wrong, and what we do about it.

Risk	What it means	Mitigation
Latency	Sequential layers pay a round trip each; internet RTT cut throughput 6x in Lumabri's gate	Batch only; place peers by RTT; batch-union and speculative drafts; Private Mesh on a LAN first
Activation privacy	Segment peers see hidden states, expert peers see expert inputs; not the prompt, but a leak	Attested tiers, private pools, customer-chosen countries; say it plainly in every doc
Verification cost	Replica re-execution is the only check today	Price the buffer; proof-of-inference on the research track
Windows and phones	Most idle desktops run Windows; no phone path in Colibri	Fund upstream Windows certification; phones stay in 1.0
Upstream dependency	Two fast-moving codebases we do not own	Pin commits, adapter layer, contribute back, never fork the engine
Supply bootstrapping	A swarm with no peers is a tracker	Private Mesh needs none; community opens on the release gate

Five milestones, each ends with a number.

1

M1 · WAN proof

Sep 2026

Tracker on GCP, one peer on GCP, one peer behind a home NAT. Real model. tok/s, per-hop RTT, snapshot on peer loss.

2

M2 · Ayni adapter

Oct

A Mesh route is a provider class: quote, accept, run, collect unchanged. Metered earnings per peer. MCP tools work as is.

3

M3 · Pilot judgment stage

Oct

The CISO pilot's flagged 2% judged on the swarm. Price per correct answer next to hosted 70B, measured on labelled lines.

4

M4 · Private Mesh installer

Nov

One installer for a team's machines, Windows included, with attestation, metering and audit. First paying private mesh.

5

M5 · Community swarm

Q1 2027

Opens when three donors beat one machine by 2x on the release gate. GPU peers, proof-of-inference spike.

Kill criteria, stated now: if M1 shows under 1 tok/s across a real WAN with batching, or M3 cannot beat hosted 70B on price per correct answer, Mesh stays a Private Mesh product and the community swarm waits for GPU peers.

What Mesh needs in the next 90 days.

10

peer machines with 32 GB+ RAM and NVMe, on loan from the community, for M1 and M5

2

pilot customers with a batch judgment workload and labelled data: one security team, one regulated SMB

1

decision: fund upstream Windows certification, or accept Linux and macOS supply for a year

- Ayni 1.0 keeps shipping: phones, the MCP server, the CISO pilot's mechanical stage. Mesh is a second track with its own repo and GCP project, so neither slows the other.
- Everything Mesh publishes is measured: throughput, price per correct answer, what peers can see. The upstream projects earned 30,000 stars that way; we will earn customers the same way.

github.com/Ayni-ai/mesh · ayni-ai.com · hello@ayni-ai.com